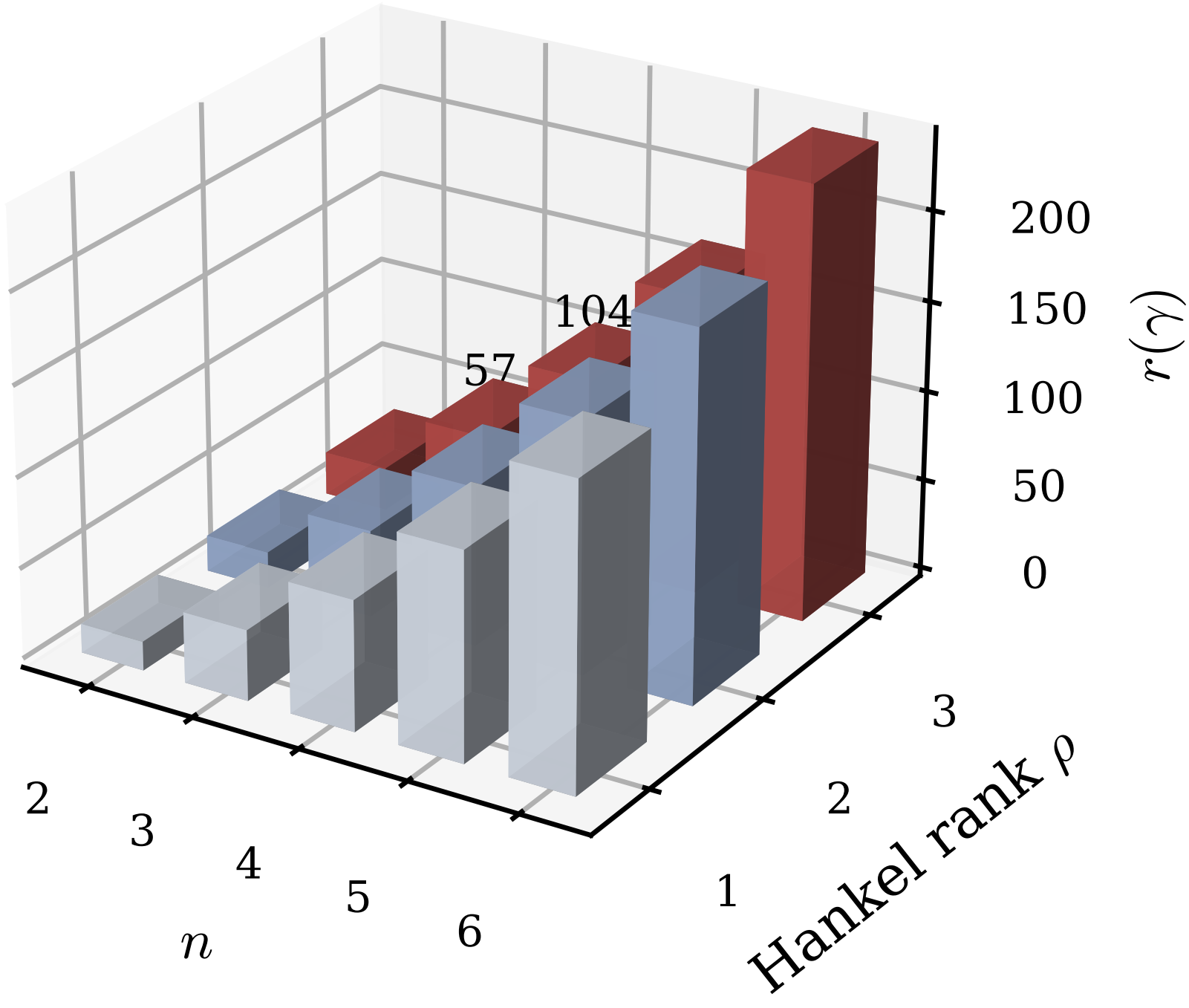$r(\gamma) = (4+\rho)n^2 - 2n$: the rank the criterion demands